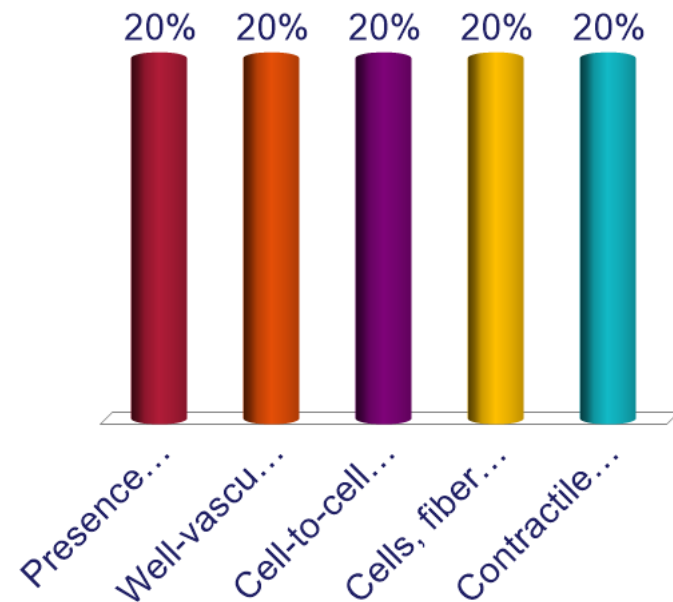


A 14-year-old boy is brought to the physician by his mother because of a 2-day history of anterior, parasternal chest pain that is exacerbated by coughing and deep breathing movements. He is recovering from an upper respiratory tract infection that presented with sore throat and frequent coughing. Physical examination shows bilateral tenderness on palpation of the costochondral regions of ribs 2 - 4. The physician suspects costochondritis (inflammation of the costal cartilage at the articulation of the ribs and sternum). Which of the following best characterizes the inflamed tissue type in this patient?

- A. Presence of a basement membrane
- B. Well-vascularized
- C. Cell-to-cell junctional complexes
- D. Cells, fibers, and ground substance
- E. Contractile proteins





**Basic Principles of Medicine 1**  
**Module: Musculoskeletal System | Lecture 18**

# **Histology of Cartilage (Flipped Class)**

**Dr. Ali Drigo**

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# Required Reading

- Histology – A Text and Atlas  
Pawlina 9<sup>th</sup> Edition  
Chapter 7: Cartilage pages

[Cartilage | Histology: A Text and Atlas: With Correlated Cell and Molecular Biology, 9e | Premium Basic Sciences | Health Library \(oclc.org\)](#)

- Pre-reading Assignment (Sakai Resources)

# Objectives

| State the function(s) of hyaline cartilage.                                                         | SOM.MK.I.BPM1.2.MSK.3.HCB.1025 |
|-----------------------------------------------------------------------------------------------------|--------------------------------|
| Describe the histological structure of hyaline cartilage.                                           | SOM.MK.I.BPM1.2.MSK.3.HCB.1026 |
| Give examples of the localizations of hyaline cartilage.                                            | SOM.MK.I.BPM1.2.MSK.3.HCB.1027 |
| State the function(s) of articular cartilage                                                        | SOM.MK.I.BPM1.2.MSK.3.HCB.1028 |
| Describe the histological structure of articular cartilage.                                         | SOM.MK.I.BPM1.2.MSK.3.HCB.1029 |
| Give examples of locations of articular cartilages                                                  | SOM.MK.I.BPM1.2.MSK.3.HCB.1030 |
| State the function of elastic cartilage.                                                            | SOM.MK.I.BPM1.2.MSK.3.HCB.1031 |
| Describe the microanatomical structure of elastic cartilage.                                        | SOM.MK.I.BPM1.2.MSK.3.HCB.1032 |
| Give examples of locations of elastic cartilage.                                                    | SOM.MK.I.BPM1.2.MSK.3.HCB.1033 |
| State the function of fibrocartilage.                                                               | SOM.MK.I.BPM1.2.MSK.3.HCB.1034 |
| Describe the microanatomical structure of fibrocartilage.                                           | SOM.MK.I.BPM1.2.MSK.3.HCB.1035 |
| Give examples of locations of fibrocartilage.                                                       | SOM.MK.I.BPM1.2.MSK.3.HCB.1036 |
| Discuss osteoarthritis and relate the histological changes in cartilage with clinical presentation. | SOM.MK.I.BPM1.2.MSK.3.HCB.1037 |

# Clinical Case 1

A 62-year-old man comes to the physician complaining of worsening pain and pressure in his upper arm for the past 2 months. The patient states that he has felt dull pain in his right shoulder that gets worse with activity.

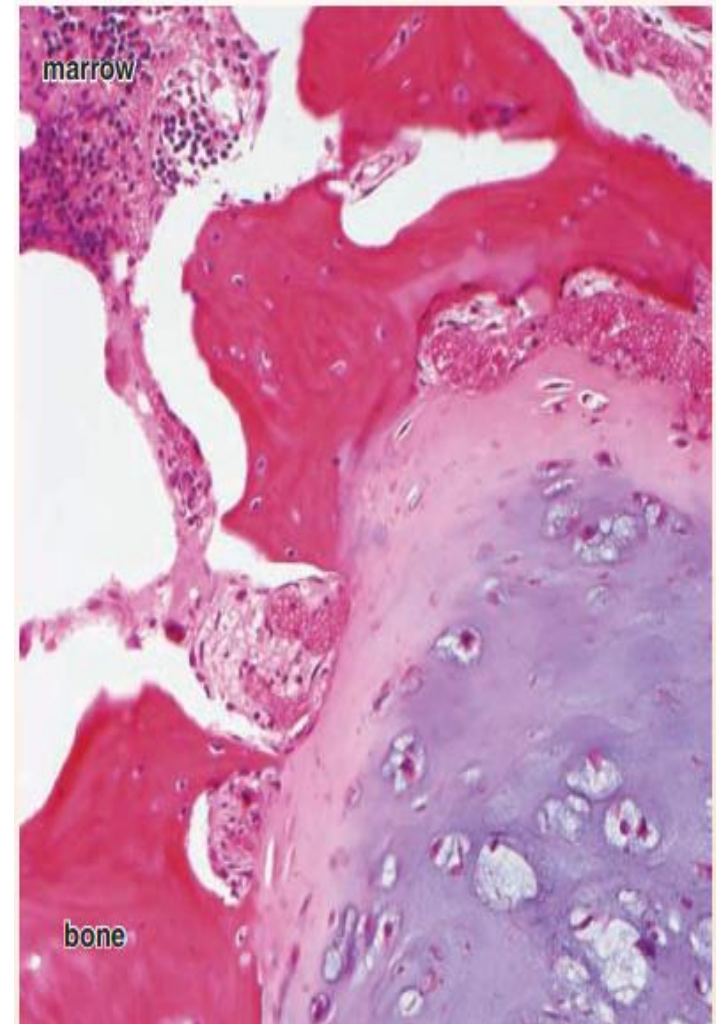
Physical examination shows tenderness and slight swelling of the proximal right humerus. Biopsy shows that the tumor is comprised of tissue similar to the tissue normally found within epiphyseal growth plate. The patient is diagnosed with a **chondrosarcoma**, a malignant tumor that secretes a cartilaginous matrix.

*Which of the major types of cartilage typically constitutes the epiphyseal growth plate?*

*Give examples of other locations in the body where we can find this tissue.*

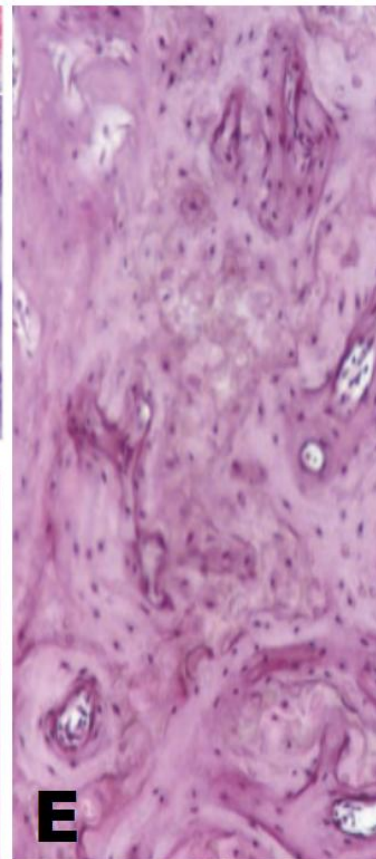
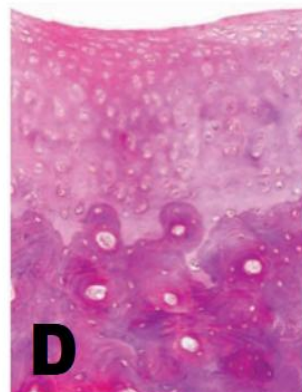
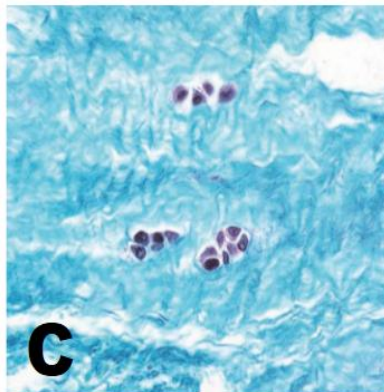
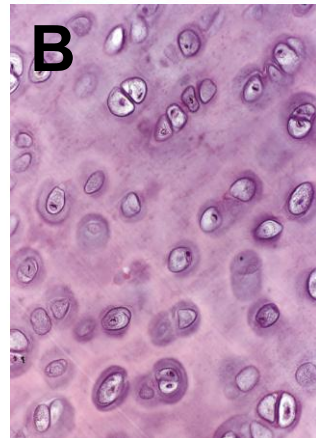
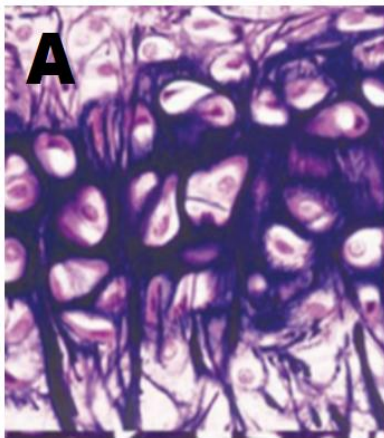
*What are some of the histological characteristics of this tissue?*

*How do these characteristics contribute to its functions?*



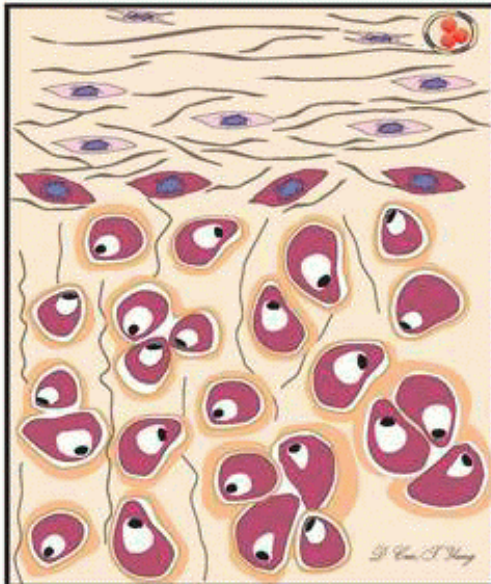
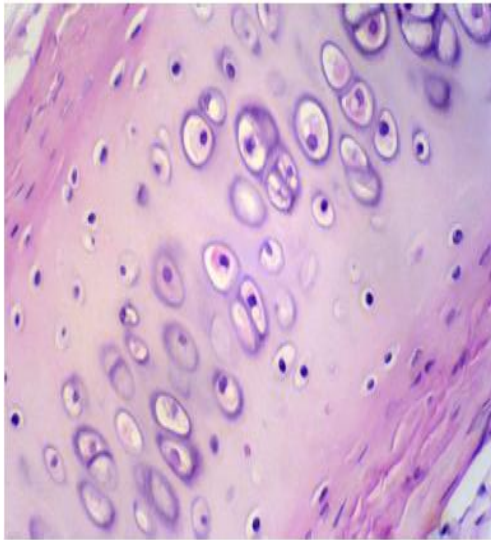
# Q1: Which of the following tissues is most likely normally found in the epiphyseal plate?

- A. A
- B. **B**
- C. C
- D. D
- E. E

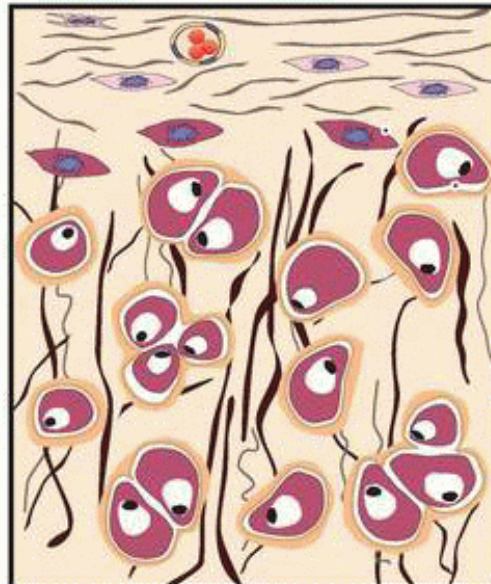
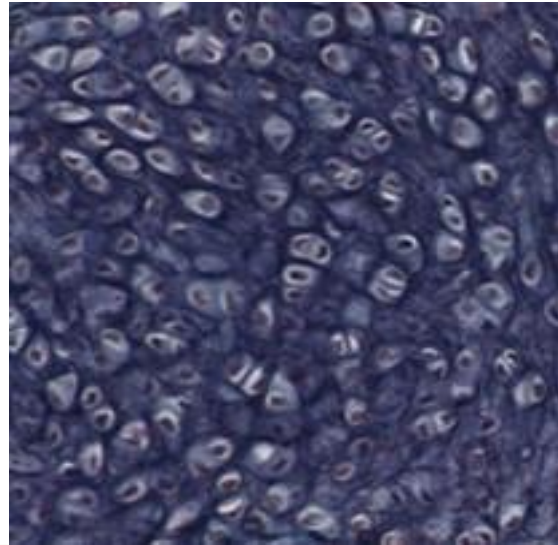


# Types of Cartilage

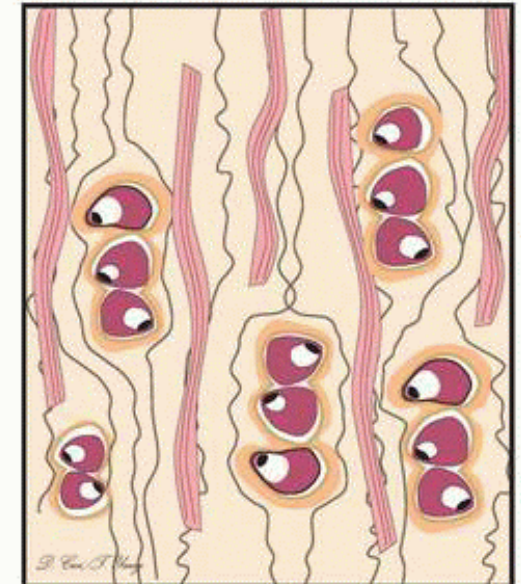
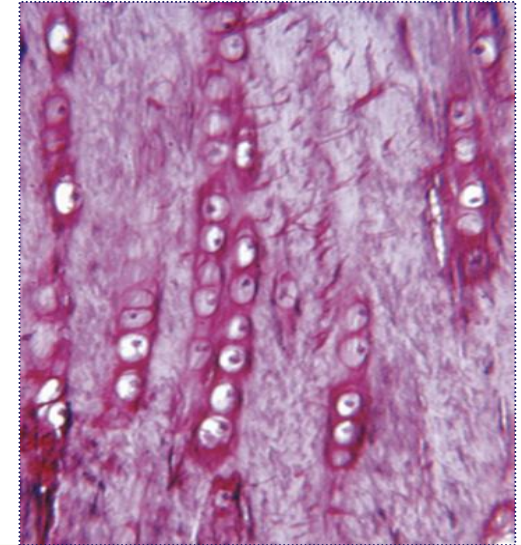
## Hyaline Cartilage



## Elastic Cartilage

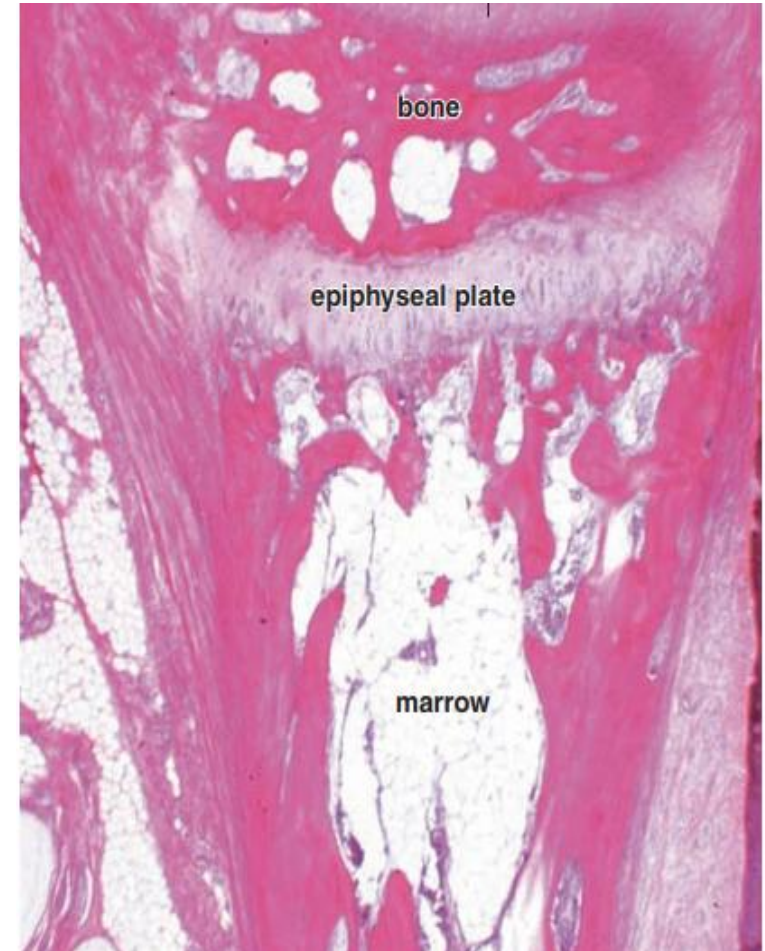
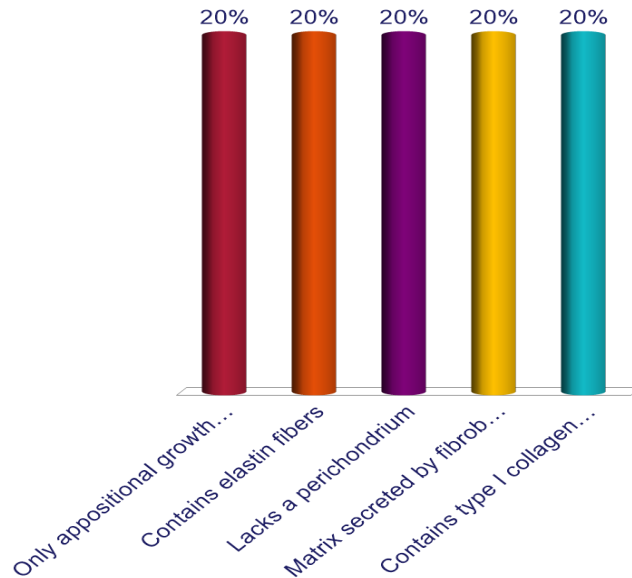


## Fibrocartilage



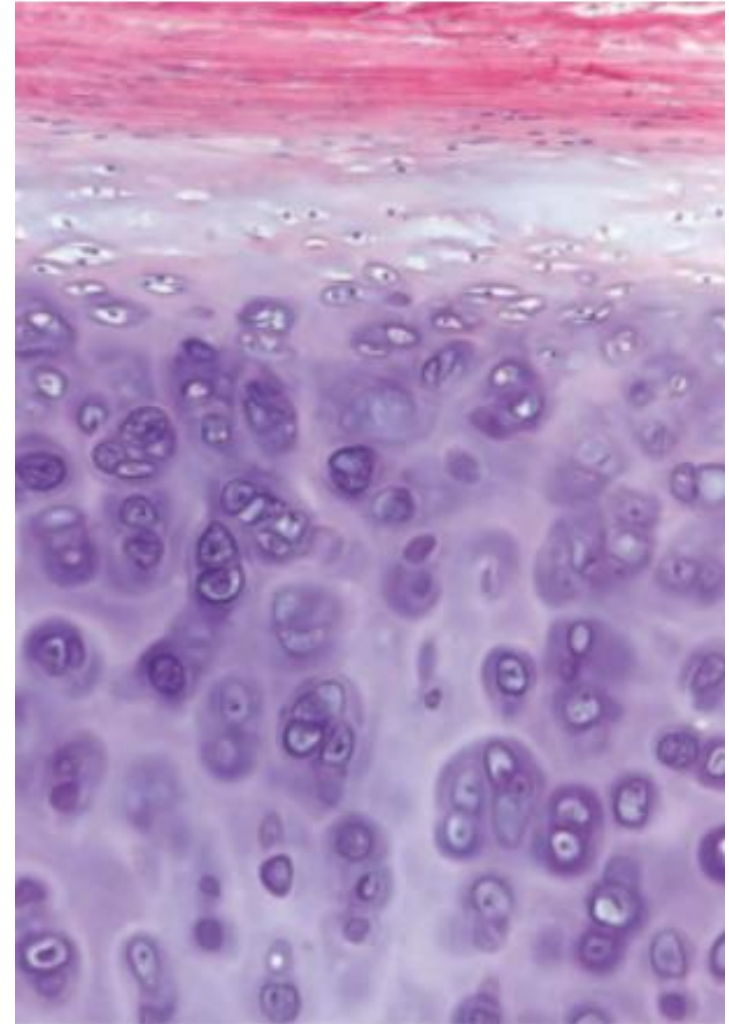
## Q2: Which of the following statements best characterizes the tissue normally found in the epiphyseal plate?

- A. Only appositional growth is seen
- B. Contains elastin fibers
- C. Lacks a perichondrium
- D. Matrix secreted by fibroblast
- E. Contains type I collagen fibers



# Hyaline cartilage

- **Histogenesis:**  
Mesenchymal cells
- **Cells:**  
**Chondroblasts** & chondrocytes
- **Extracellular matrix:**  
Type II collagen fibers  
GAGs - hyaluronic acid and glycoproteins
- **Growth:**  
Appositional & interstitial growth
- **Degeneration:**  
Does not readily degenerate
- **Calcification:**  
Calcifies – bone formation & aging



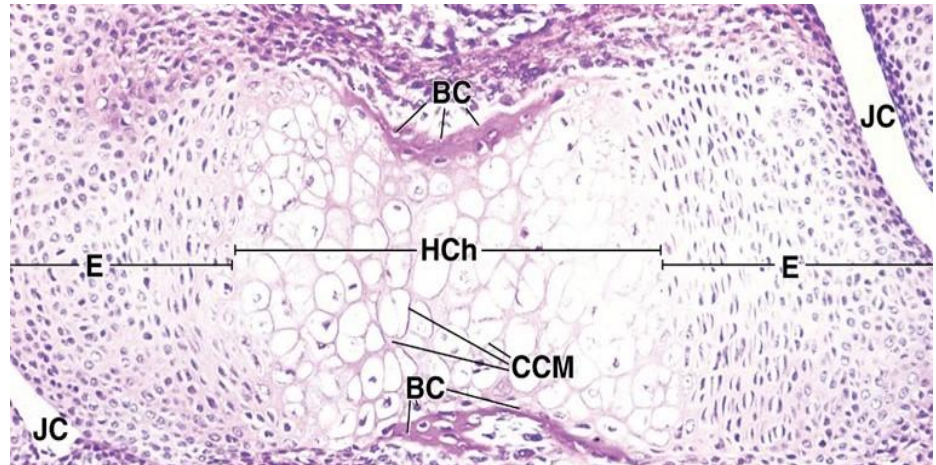
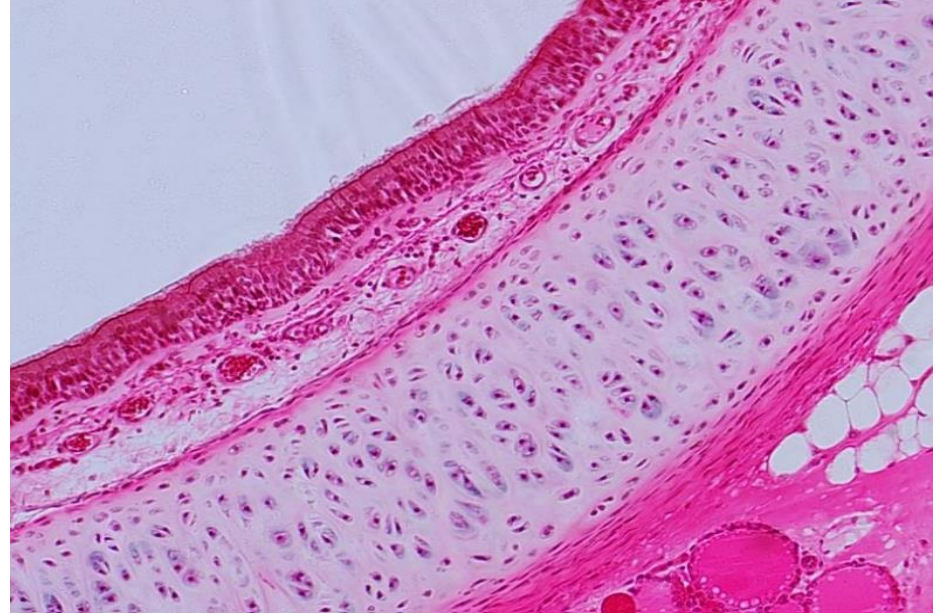
# Hyaline cartilage

## Functions:

- Supportive
- Cushioning
- Shock absorber
- Growth plate
- Model for bone formation

## Histology

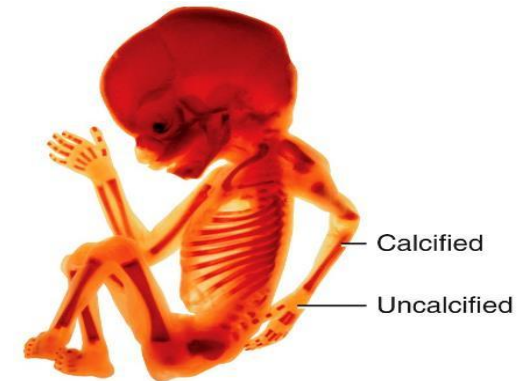
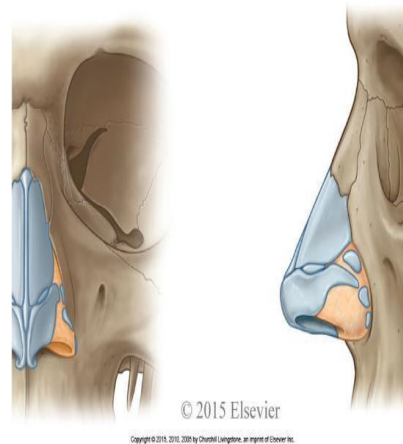
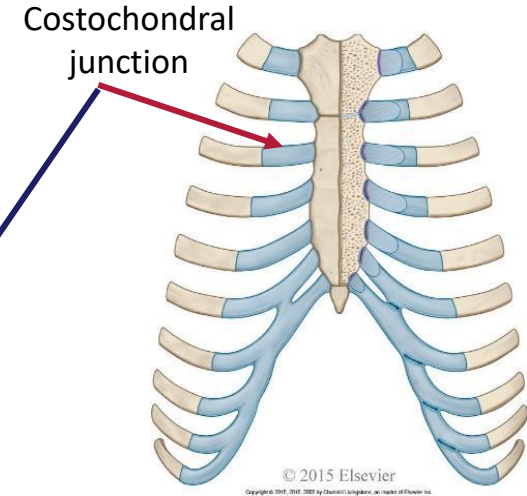
- Matrix has a “glassy” appearance because fibers and ground substance has the same refractive index
- ECM correlates to function!
  - Collagen
  - Sulfated groups



# Hyaline cartilage

## Locations in the body:

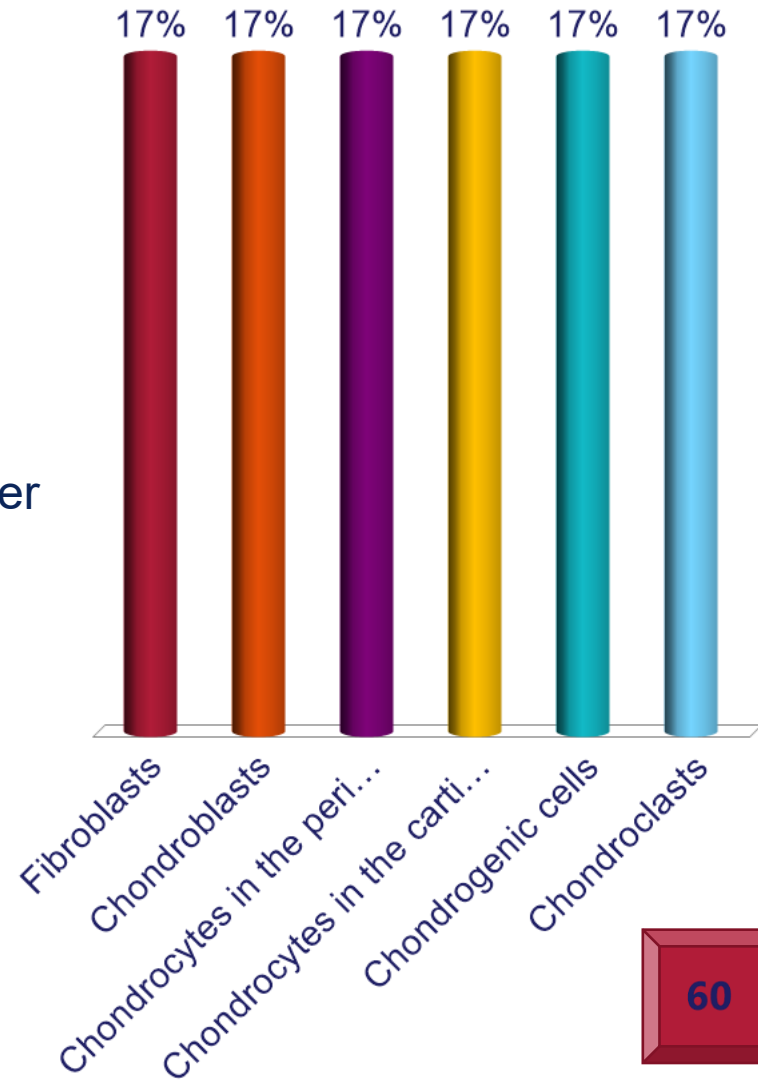
- Nose (cartilaginous portion)
- Tracheal and bronchial rings
- Laryngeal cartilages
- Costal cartilages
- Articular surfaces of long bones
- Epiphyseal growth plate
- Fetal skeleton



(b) Twelve-week fetus. The red areas represent bones that are forming (calcified). Clear areas represent cartilage (uncalcified).

# Q3: Which of the following cells are most likely responsible for appositional growth in the previously described type of cartilage (hyaline)?

- A. Fibroblasts
- B. Chondroblasts
- C. Chondrocytes in the perichondrium
- D. Chondrocytes in the cartilage matrix center
- E. Chondrogenic cells
- F. Chondroclasts

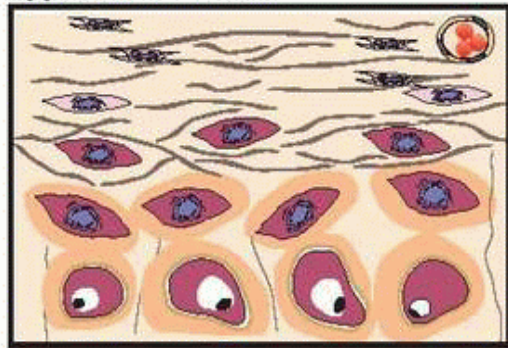


# Types of growths seen in hyaline cartilage:

## Appositional Growth

- occurs from chondrogenic cells in the perichondrium differentiating into chondroblasts, forming a new layer of cartilage around the periphery of the existing cartilage.
  - This increases the width or thickness of the cartilage

Appositional Growth



### Hormonal influence on growth:

- Growth hormone
- Thyroxine
- Testosterone



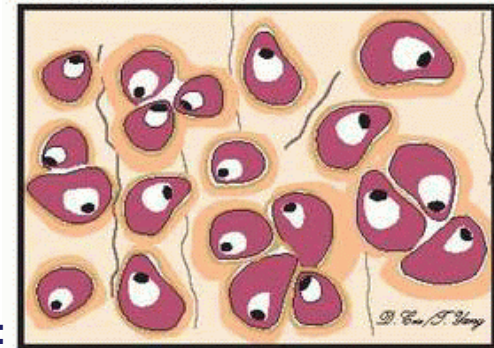
- Hydrocortisone
- Estradiol



## Interstitial Growth

- occurs only in young cartilage from cell divisions within the cartilage during the formation of isogenous groups.
  - This increases the length of the cartilage

Interstitial Growth



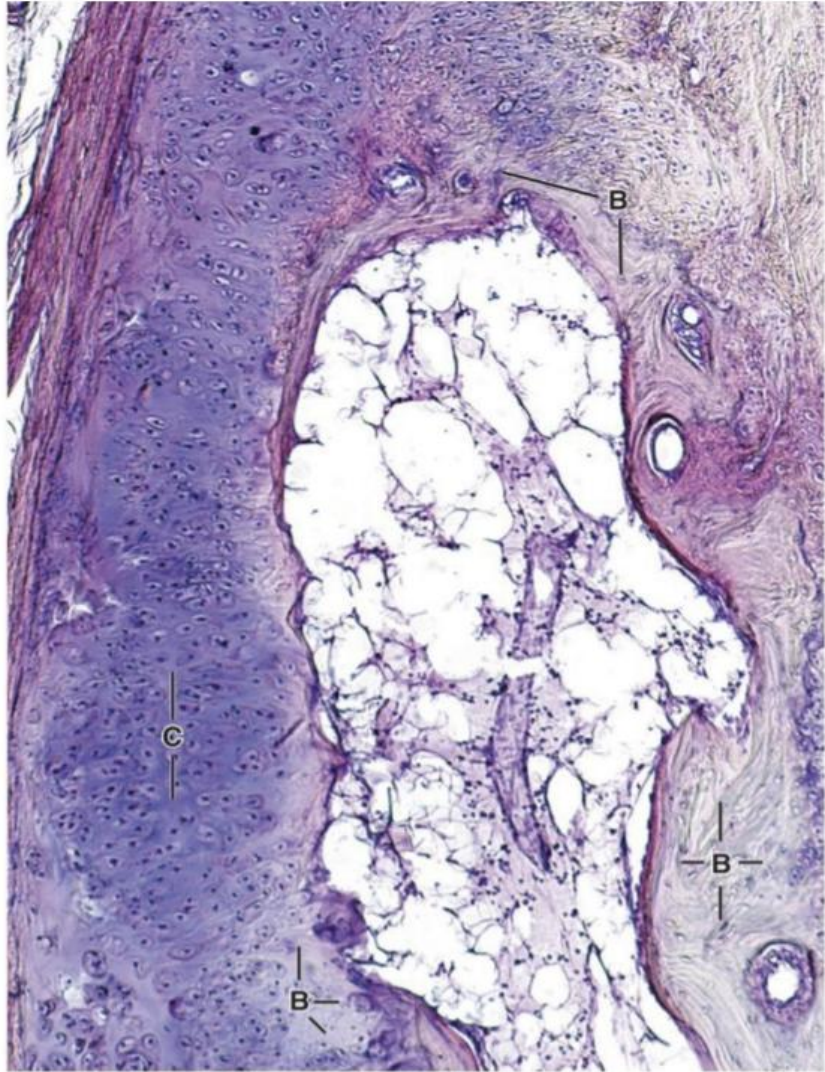
# Hyaline cartilage

- **Regeneration**

Hyaline cartilage regenerates **very poorly** and often the perichondrium forms scar tissue

- **Calcification:**

In old age hyaline cartilage can become calcified



Micrograph of tracheal rings with showing ageing related calcification of cartilage (C – cartilage, B – bone)

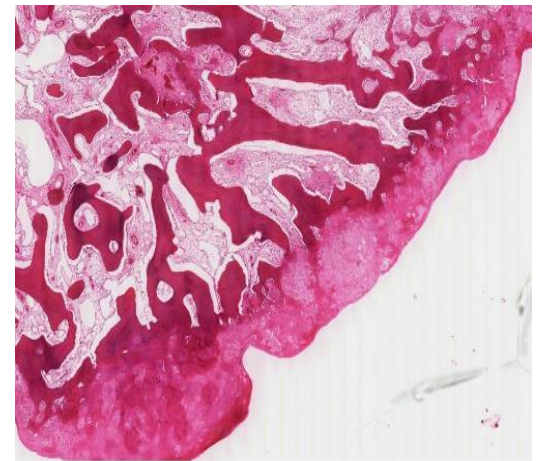
## Clinical Case 2

A 58-year-old woman comes to the physician because of progressively worsening pain and stiffness in her knees for the past 6 months. She states the pain gets worse with physical activity and tends to improve with rest.

Physical examination shows tenderness on palpation of the medial aspect of her knees. Radiographic imaging shows bilateral joint space narrowing and osteophyte formation at the articular surfaces of the knee joints.

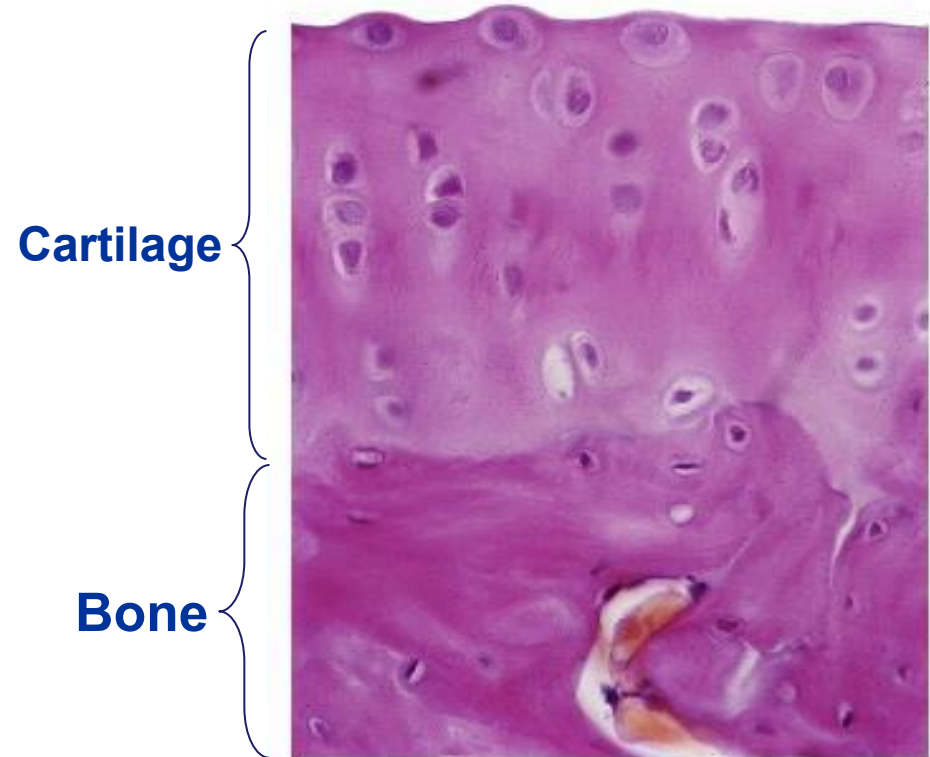
*Describe the zones of the specialized tissue that normally lines the articular surfaces of long bones.*

*How do the cells of this tissue receive nutrients?*



# Articular cartilage

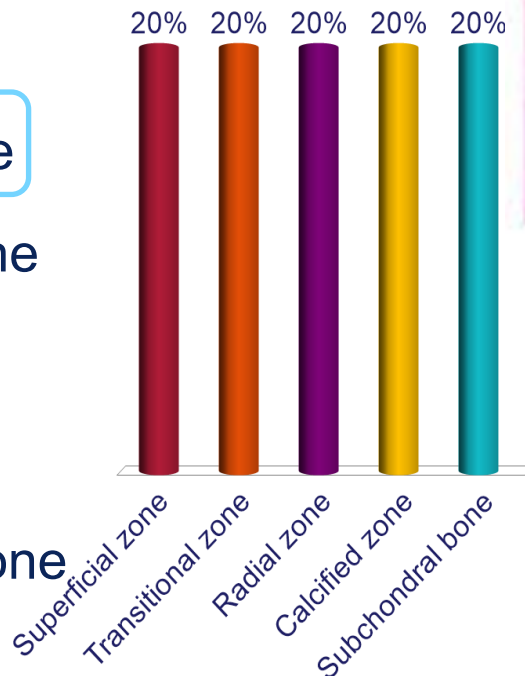
- A type of hyaline cartilage found at articular surfaces
  - **does not have perichondrium.**
- It is in contact with articular surface of bone on opposite side.
- Remnant of original cartilage template of developing bone.
- Persists throughout adult life.



**Q4: A tissue sample from articular cartilage shows complete erosion of the zone that is typically characterized by elongated and flattened chondrocytes surrounded by type II collagen fibrils. Which of the following regions most likely corresponds to the affected zone?**



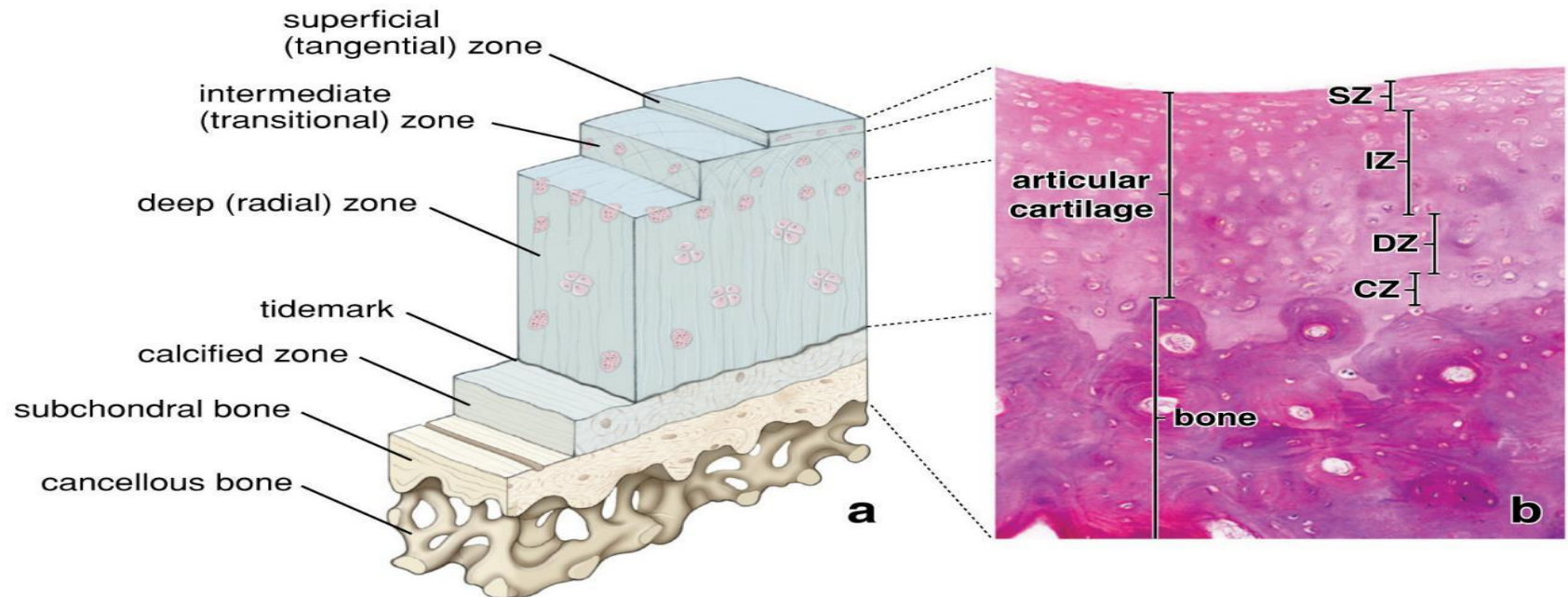
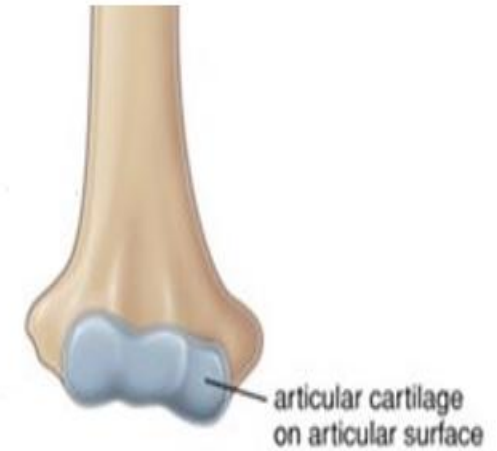
- A. Superficial zone
- B. Transitional zone
- C. Radial zone
- D. Calcified zone
- E. Subchondral bone



# Articular Cartilage

## Various regions of articular cartilage:

1. Tangential layer
2. Transitional layer
3. Radial layer
4. Calcified cartilage



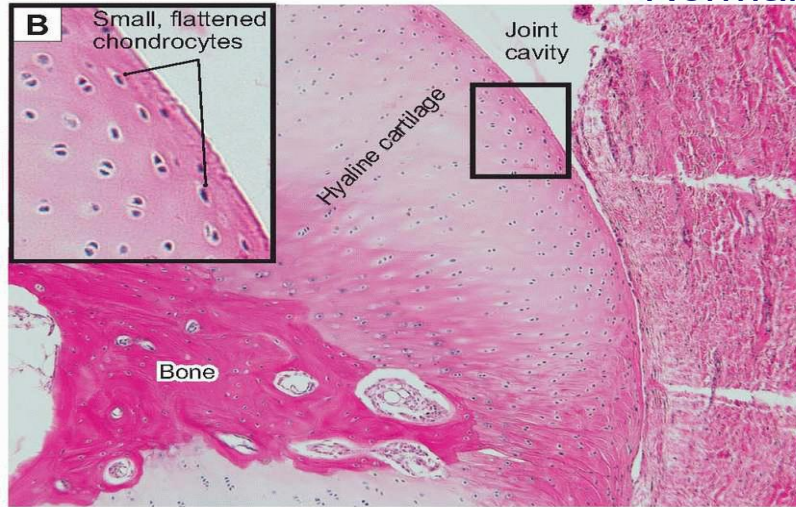
# Osteoarthritis

- A degenerative joint disease that affects the weight bearing joints (e.g., hips, knees, lower vertebrae).
- Related to aging and injury.
- Decreased proteoglycan content, which results in reduction in intercellular water content in the cartilage matrix.
- Patients present with progressive reduction of mobility and increased pain with joint movement.

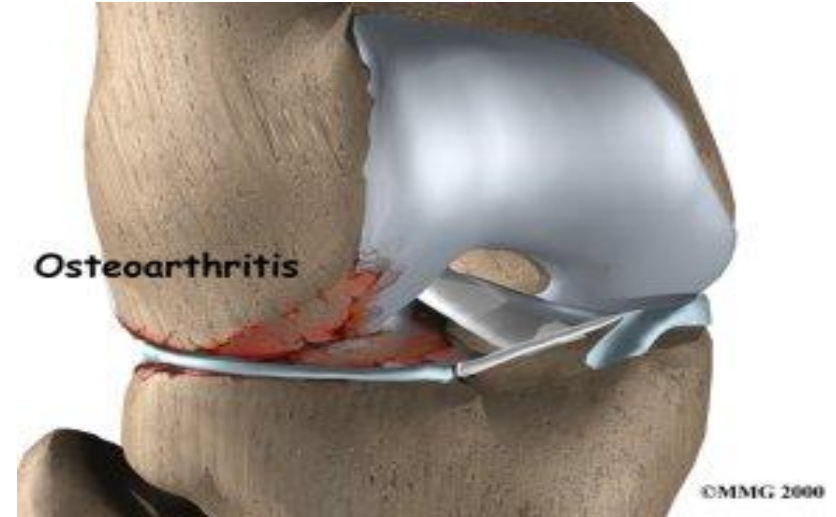
## Clinical correlate!

## Compare articular cartilages

## Normal



## Osteoarthritis



## Osteoarthritis



## Osteoarthritis



# Clinical Case 3

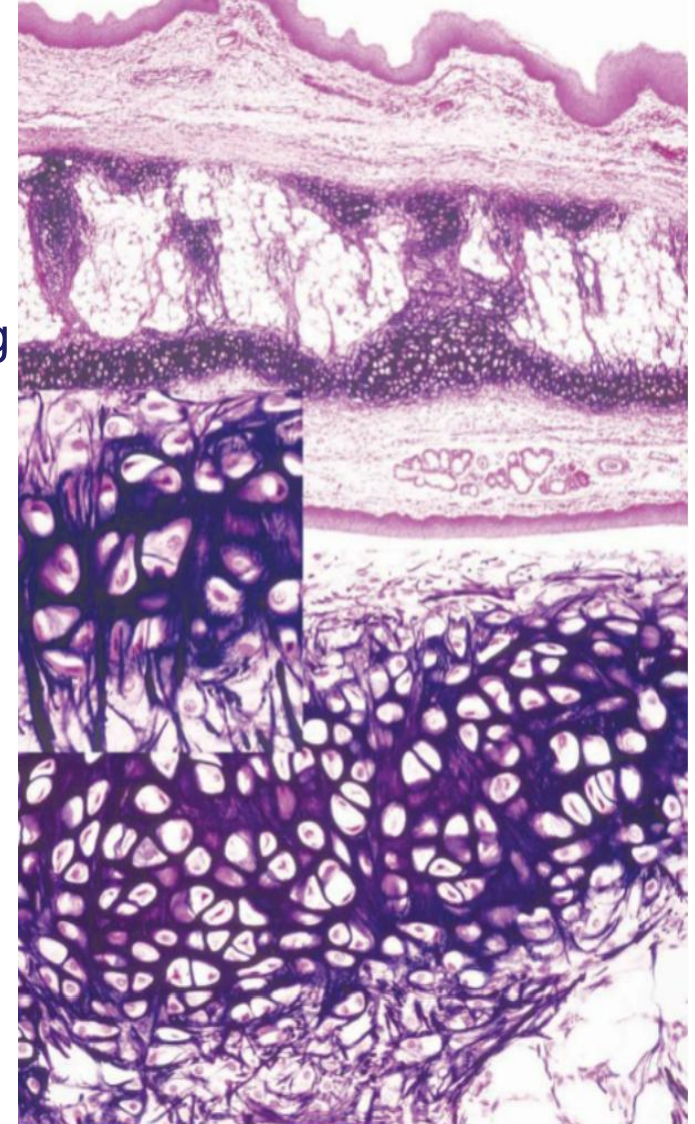
A 3-year-old boy is brought to the emergency department by his parents because of a 4-hour history of hoarseness, severe sore throat, and a high fever.

Physical examination shows an acutely ill child that is leaning forward with drool coming out of his mouth. Direct observation of the larynx reveals a swollen and erythematous epiglottitis.

He was diagnosed with **bacterial epiglottitis** and administered treatment and antibiotics according to established guidelines.

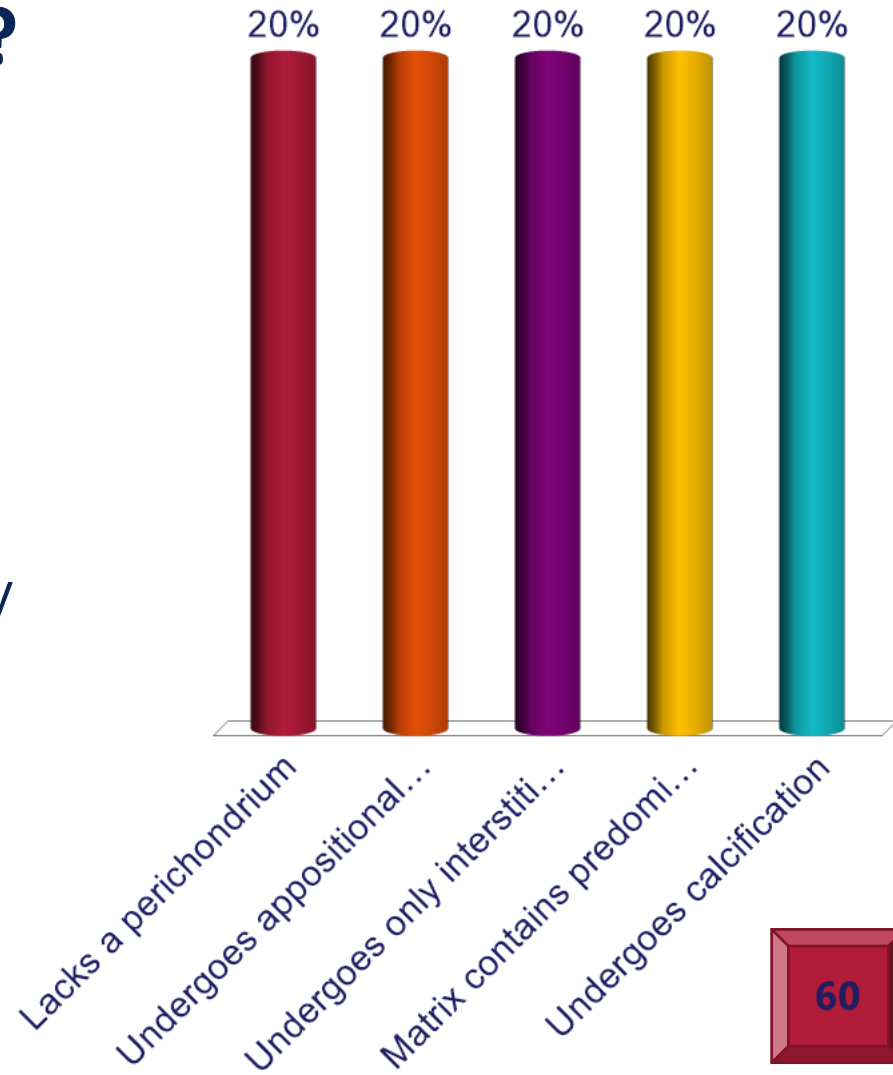
*What type of cartilage forms the core of the inflamed structure (epiglottis)?*

*How is this type of cartilage similar to hyaline cartilage? How are they different? (Compare composition, function, location, etc.).*



## Q5: Which of the following is a characteristic feature of the cartilage in the swollen structure described in this patient?

- A. Lacks a perichondrium
- B. Undergoes appositional and interstitial growth
- C. Undergoes only interstitial growth
- D. Matrix contains predominantly Type I collagen fibers
- E. Undergoes calcification



# Elastic cartilage

## Histogenesis:

- Mesenchymal cells

## Cells:

- Chondroblasts and chondrocytes

## Extracellular matrix:

- Type II collagen fibers & **elastic** fibers
- GAGs and glycoproteins, multi-adhesive glycoproteins

## Growth:

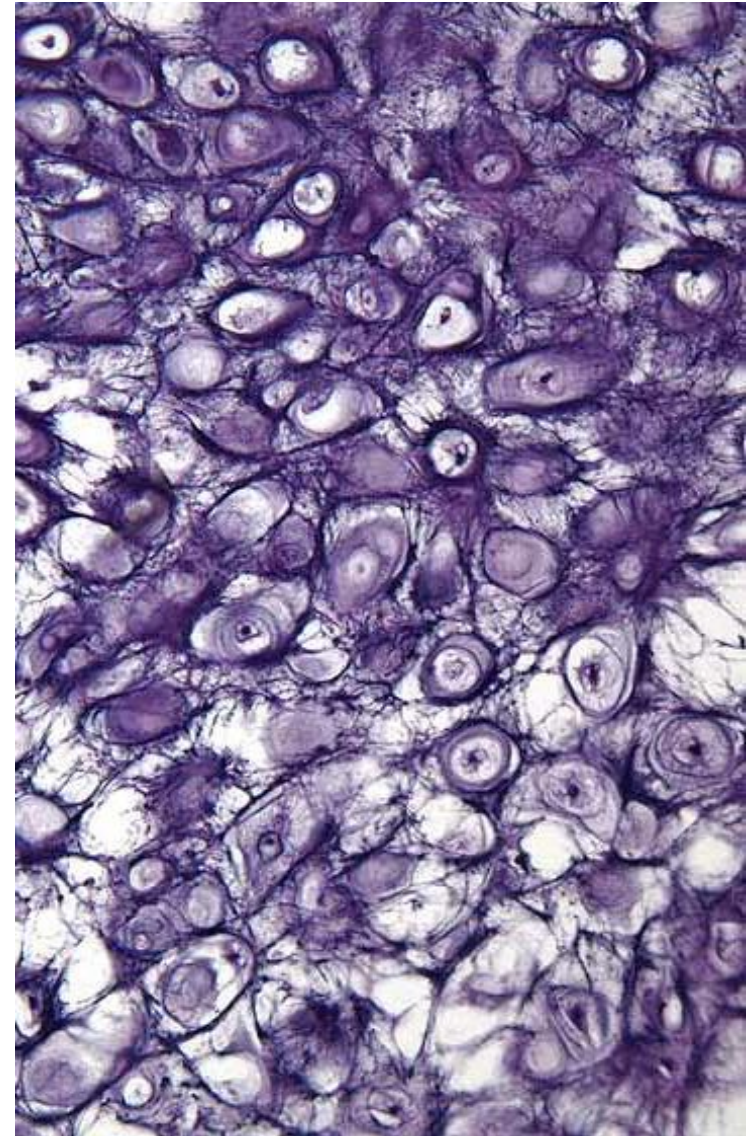
- Appositional & interstitial growth

## Degeneration:

- Does not readily degenerate

## Calcification:

- Never calcifies



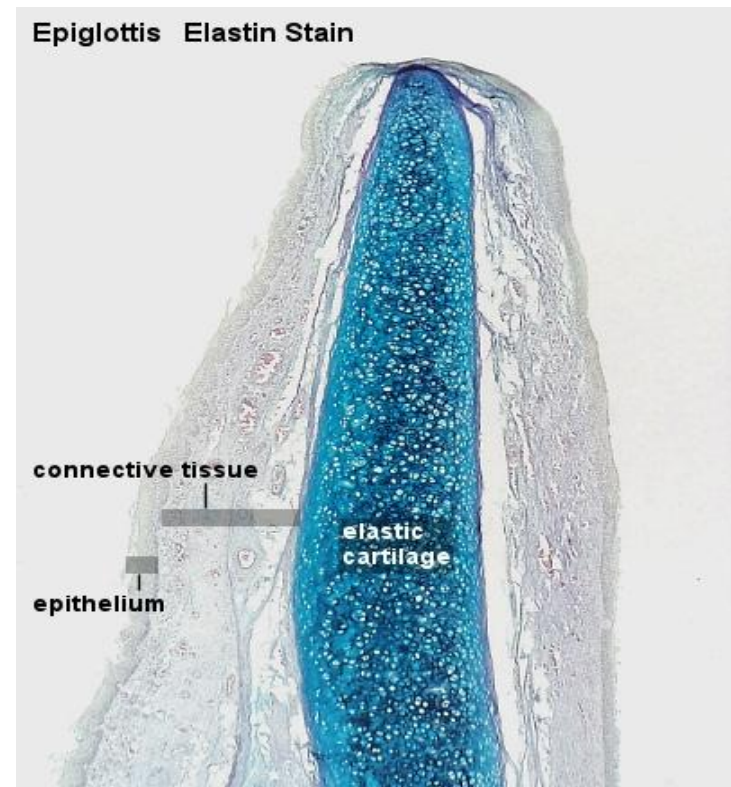
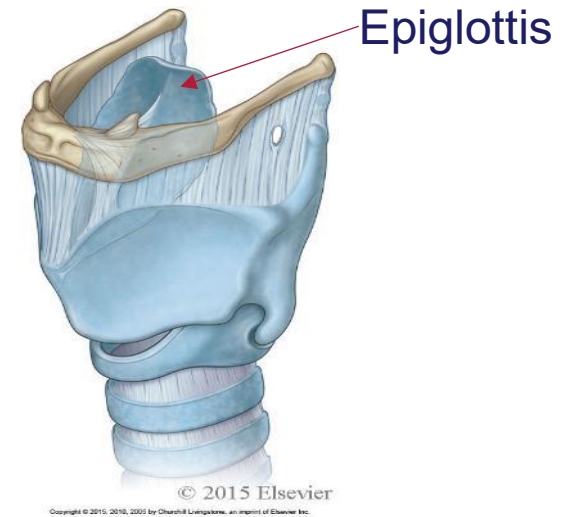
# Elastic cartilage

## Function:

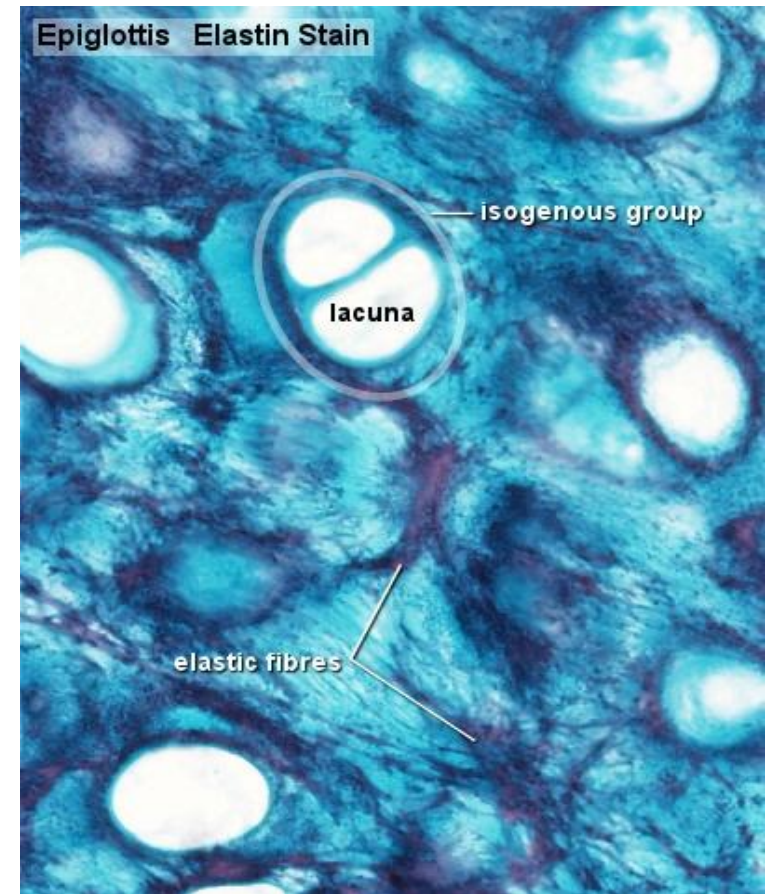
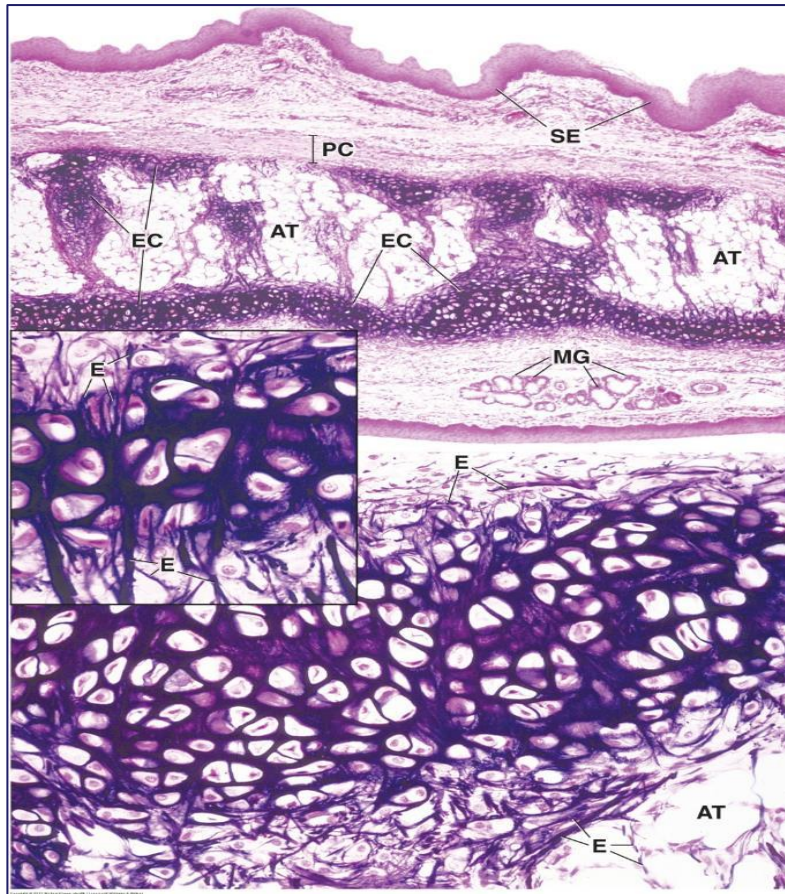
- - Support with flexibility

## Locations in the body:

- Pinna of the ear
- External acoustic meatus
- Eustachian tube
- Epiglottis
- Often found with hyaline cartilage
- **Perichondrium** present



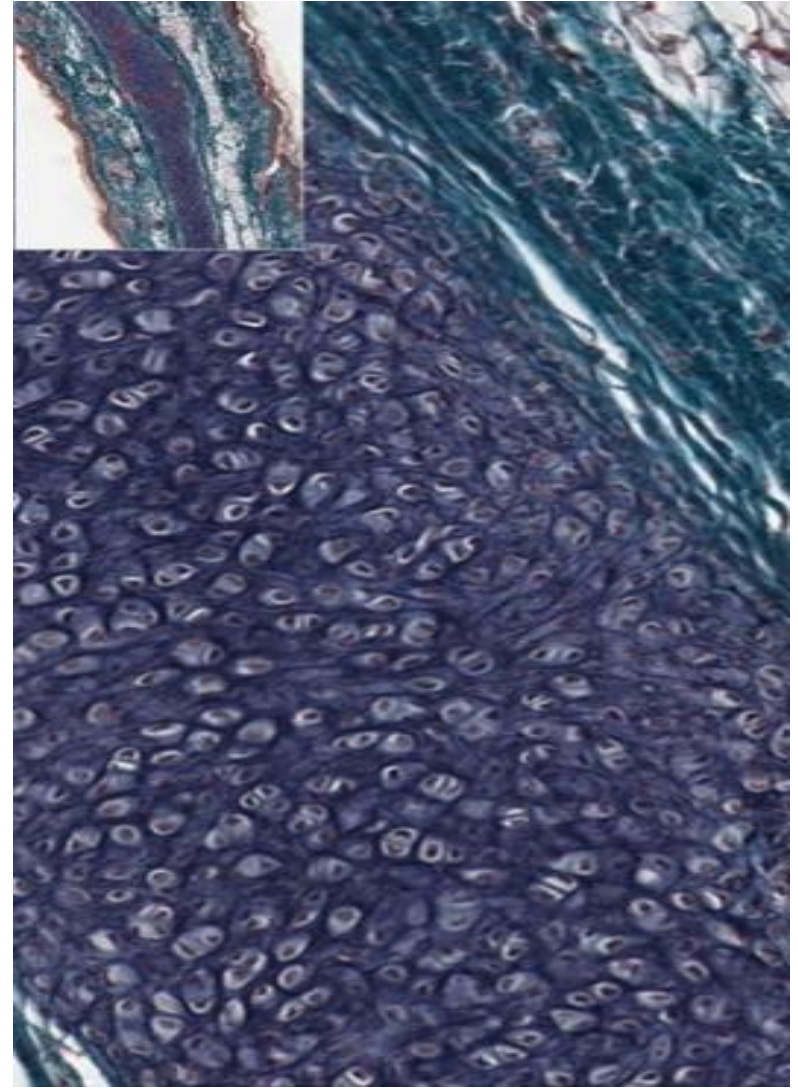
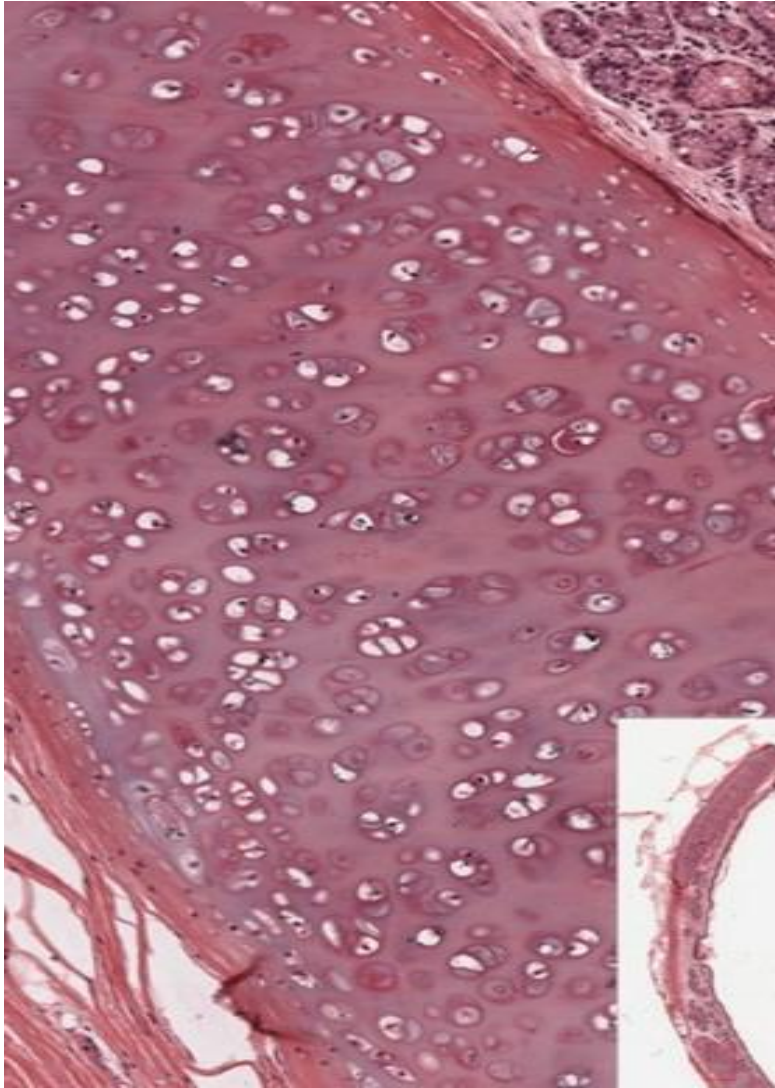
- Matrix is identical to that of hyaline cartilage, except it also contains a network of elastic fibers
  - Elastic fibers contribute a yellowish color on gross specimens



# Compare

SOM.MK.I.BPM1.2.MSK.3.HCB.1026

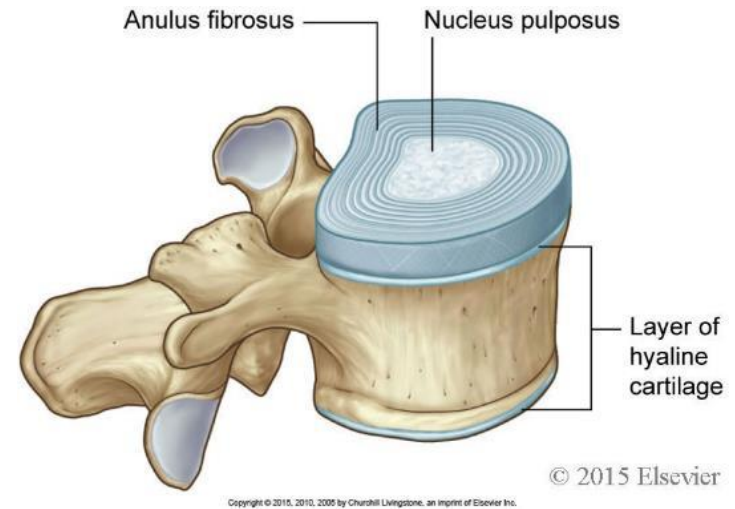
SOM.MK.I.BPM1.2.MSK.3.HCB.1032



# Clinical Case 4

A 50-year-old woman comes to the emergency room because of severe lower back pain for the past 2 days. She reports that the pain started after she attempted to pick up a heavy box at her workplace.

Physical examination shows tenderness in the lumbar region on movement and palpation. Imaging studies of the spine show an intervertebral (IV) disc herniation at the L4/5 vertebral level.

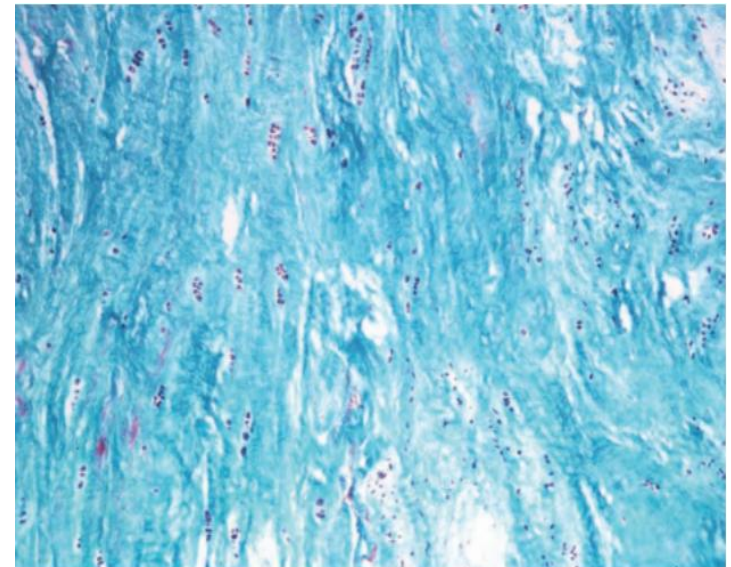


*What component of the IV disc was damaged to result in disc herniation?*

*What type of tissue is this damaged component made up of?*

*How does this tissue type differ from the other types of cartilage?*

*How does the development of this tissue explain its cellular composition?*



# Fibrocartilage

## Histogenesis:

- Mesenchymal cells

## Cells:

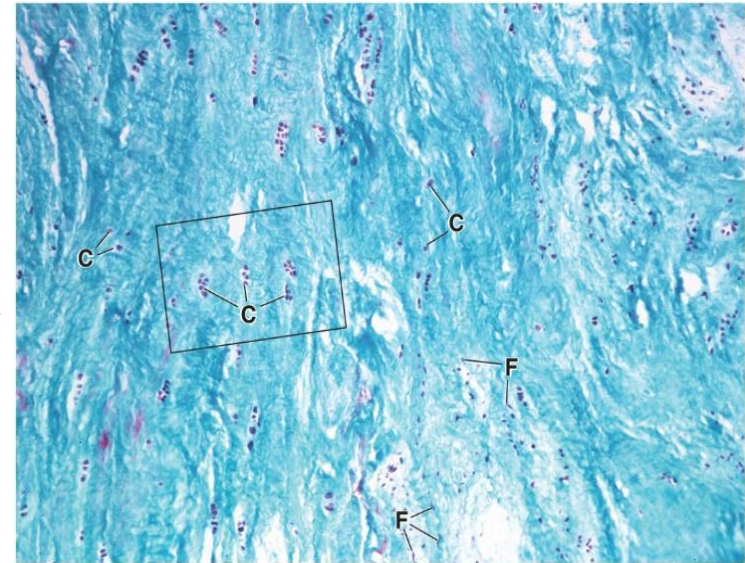
- Fibroblasts transform to chondrocytes under stress

## ECM:

- produced by chondrocytes and fibroblasts
- Predominantly **Type I** collagen fibers
- Also contains **Type II** collagen fibers
- GAGs - predominantly chondroitin and dermatan sulphates

## Growth:

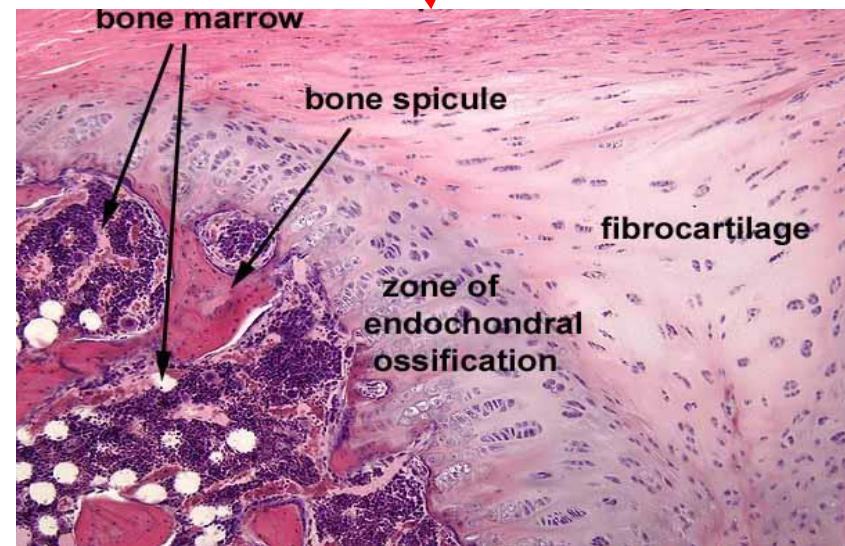
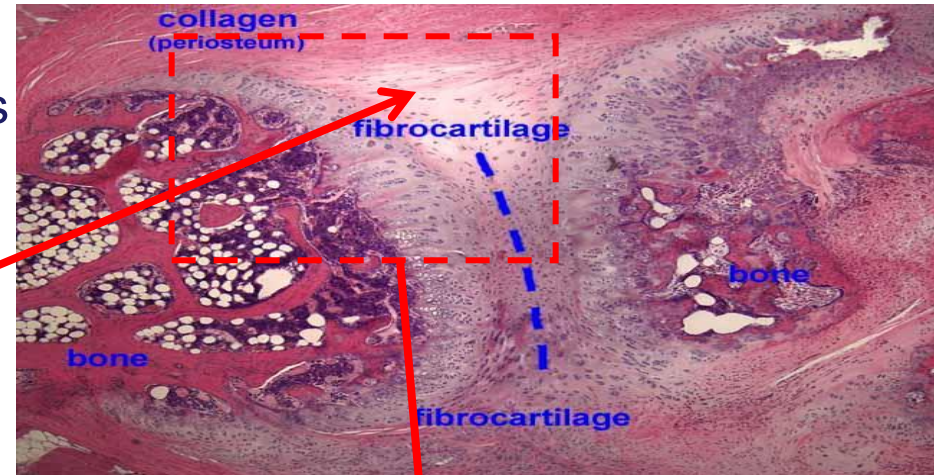
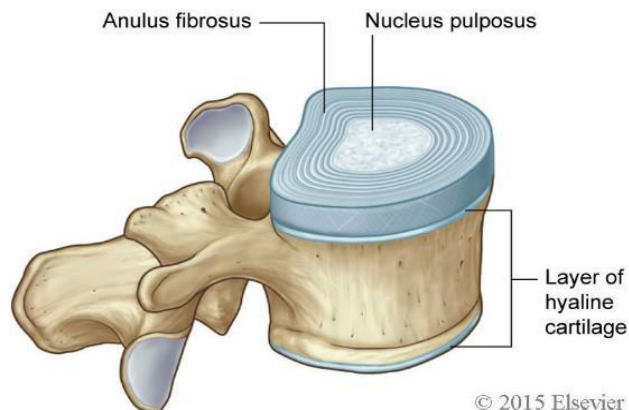
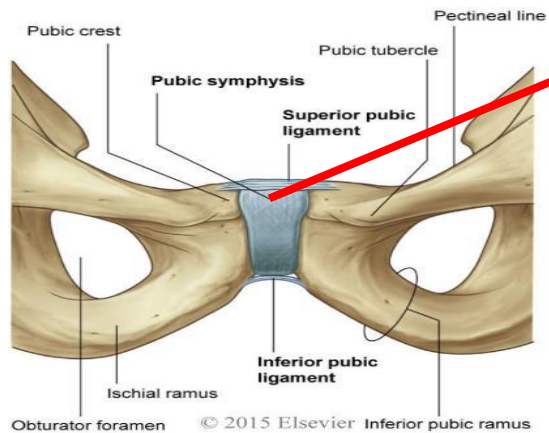
- Only interstitial growth (due to absence of perichondrium)



# Fibrocartilage

## Function

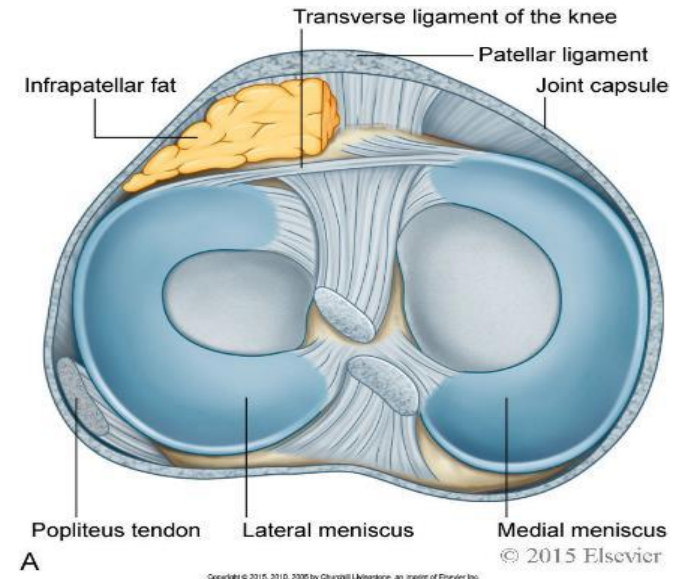
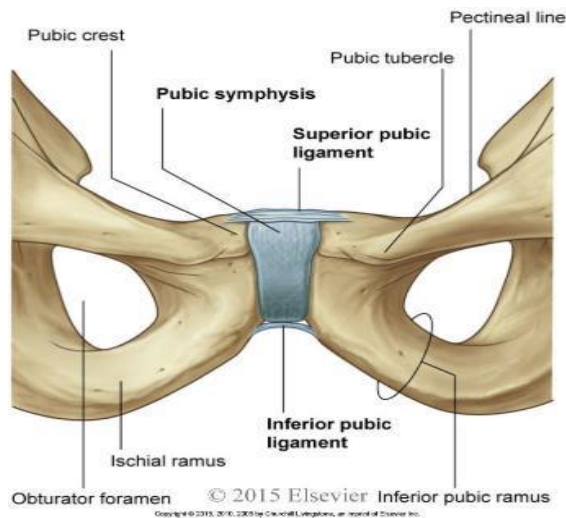
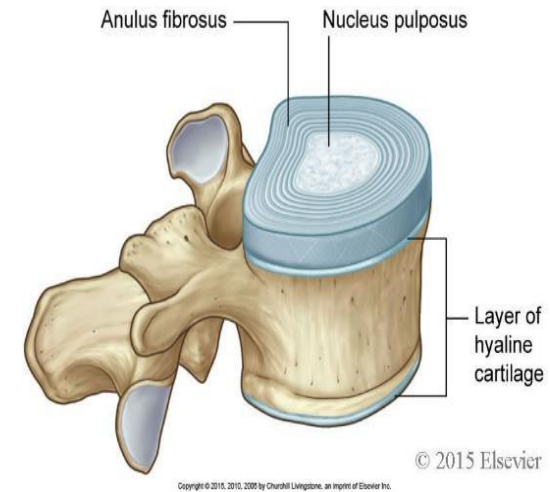
- Resists deformation under stress  
(Support and tensile strength)



# Fibrocartilage

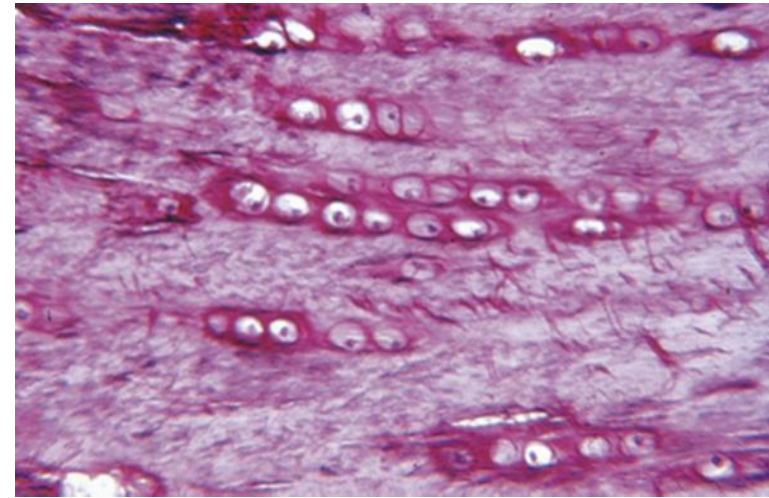
## • Locations in the body:

- Intervertebral disc (annulus fibrosus)
- At the pubic symphysis
- Medial & lateral menisci
- Articular discs
- Attachment of some tendons & ligaments to bone
- Intersections of some tendons and ligaments

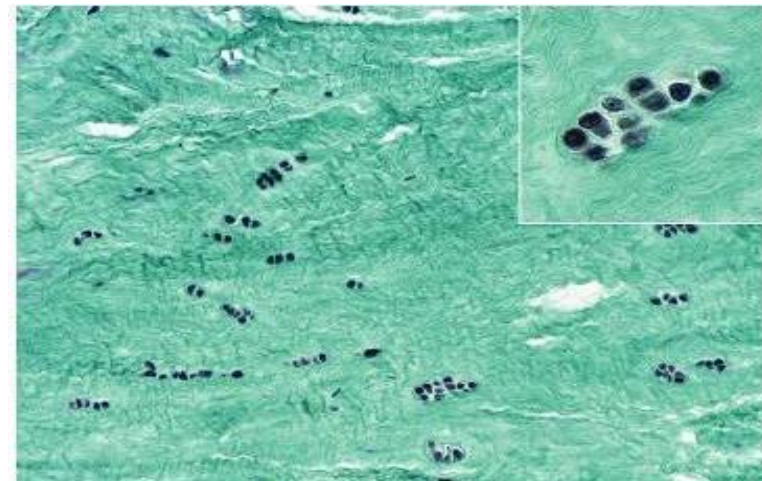


# Fibrocartilage

- Absence of perichondrium
- Matrix contains abundant type I & type II collagen fibers
  - These fibers can be seen here with H&E and trichrome staining
- *Fibers arranged perpendicular to the direction of stress*
  - Isogenous groups are in *linear fashion (longitudinal columns)* due to the compression of collagen fibers
- Can undergo calcification
- Does not occur alone, but is found in conjunction with hyaline cartilage and other fibrous tissues



**H & E stain- LM**



**Trichrome stain- LM**

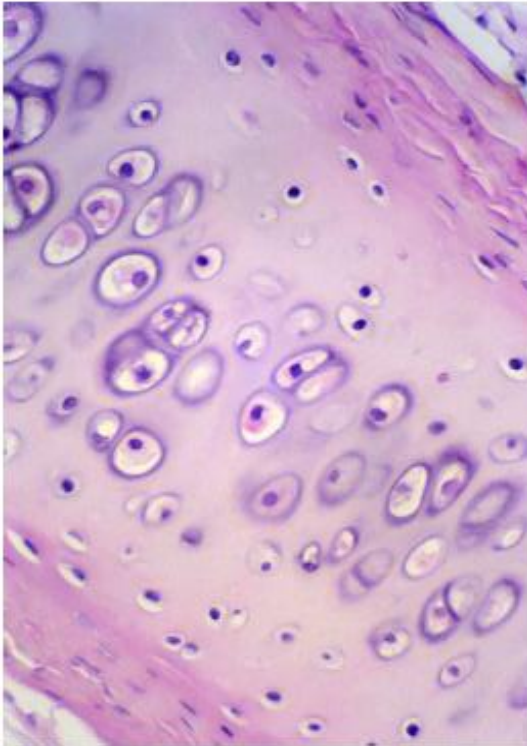


## ***Ask yourself:***

- What is cartilage?
- Name the 3 major types of cartilage and state 2 sites where each can be found.
- What type of collagen fibers are predominant in hyaline cartilage?
- What are the 2 layers of the perichondrium and what is the composition of each layer?
- How is fibrocartilage different from the other types?
- What are isogenous groups?
- What is the predominant ground substance found in cartilage?
- What are the 2 types of growths observed in hyaline cartilage?
- What is peculiar about articular surface of the long bones?
- When bone develops on cartilage model, what kind of bone formation is it? (...more on this in the 'Bone' lectures)

# Compare and contrast

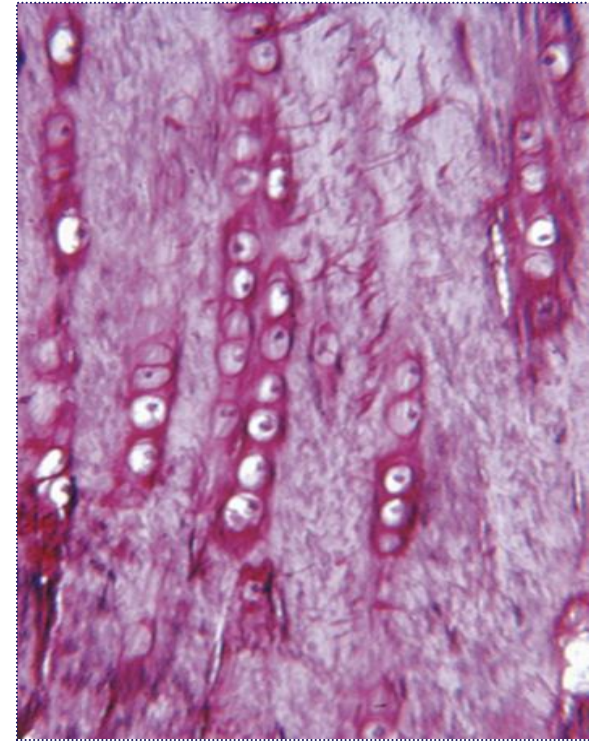
Hyaline Cartilage



Elastic Cartilage

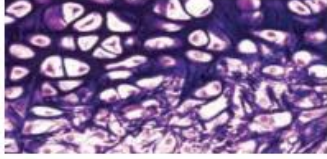


Fibrocartilage



Make a table based on :

- a) locations, b) fibers in the ECM, c) presence or absence of perichondrium
- d) types of growth e) appearance of isogenous groups.

| Features                       | Hyaline Cartilage                                                                                                                                | Elastic Cartilage                                                                   | Fibrocartilage                                                                                                                              |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Histological appearance</b> |                                                                 |  |                                                          |
| <b>Location</b>                | Tracheal and bronchial rings, Laryngeal cartilages, Costal cartilages, Articular surfaces of long bones, Epiphyseal growth plate, Fetal skeleton | Pinna of the ear<br>External acoustic meatus<br>Eustachian tube<br>Epiglottis       | Intervertebral disc, pubic symphysis, menisci, attachment of ligament to bone, intersections of some tendons and ligaments, articular discs |
| <b>Perichondrium</b>           | Yes                                                                                                                                              | Yes                                                                                 | No                                                                                                                                          |
| <b>Cell types</b>              | Chondroblasts + Chondrocytes                                                                                                                     | Chondroblasts + chondrocytes                                                        | Chondrocytes + fibroblasts                                                                                                                  |
| <b>Extracellular matrix</b>    | Type II collagen, proteoglycans                                                                                                                  | Type II collagen, elastic fibers, proteoglycans                                     | Type I and II collagen fibers, proteoglycan                                                                                                 |
| <b>Growth</b>                  | Appositional and interstitial                                                                                                                    | Appositional and interstitial                                                       | Interstitial growth only                                                                                                                    |
| <b>Calcification</b>           | Yes                                                                                                                                              | No                                                                                  | Yes                                                                                                                                         |

# Q6: Which cell types would be predominantly found in the tissue damaged to allow for the intervertebral disc herniation?

- A. Chondroblasts and chondrocytes
- B. Chondrocytes and fibroblasts
- C. Chondroblasts and fibroblasts
- D. Osteocytes and fibroblast
- E. Osteoblasts and osteocytes

